Chittagong University of Engineering & Technology

Department of Biomedical Engineering



PROJECT REPORT

Personal Rehab Assistant

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***Abstract*—This is a report on the development of a portable finger exercise device designed to continuously monitor and enhance rehabilitation activities. The device incorporates flex sensors, force-sensitive resistors, interfaced with an ESP32 microcontroller to collect data on arm movements. This data is crucial for therapists to fine-tune treatment plans and provide patients with an effective home-based rehabilitation solution. Recent findings have shown the potential of home-based rehabilitation. The system integrates various sensors, including flex sensors, force-sensitive resistors, which are interfaced with an microcontroller. These sensors capture data related to fingers movements, which is essential for therapists to tailor treatment plans and for patients to track their progress. The device aims to bridge the gap between costly and complex rehabilitation technologies and the need for accessible, home-based solutions. The continuous monitoring capabilities of the device offer valuable insights into patients' progress, improving the quality of rehabilitation programs. This work aligns with recent advancements in wearable technology and sensor systems for healthcare applications, emphasizing the importance of accessible rehabilitation solutions.**

Keywords- ESP 32, flex sensors, force-sensitive resistors,Servo motor MG996R,Motor Driver PCA9685,Veroboard

I. Introduction

Paralysis is a leading cause of disability worldwide, affecting millions of people each year. One of the most common consequences of stroke is the loss of motor function, particularly in the hands. This loss of function can severely impact a person's quality of life and limit their Rehabilitation is a critical component of stroke recovery, and there is a need for innovative solutions to aid in this process. One such solution is the developments of a smart glove, a wearable device that can help stroke survivors regain the use of their hands.

The smart glove is a wearable device that uses sensors to detect hand movements and provide feedback to the use . These exercises are designed to improve range of motion, strength, and dexterity in the hand. The smart glove is non-invasive, cost-effective, and easy to use, making it an attractive solution for stroke survivors.The problem addressed by this project is the development of a smart glove for stroke rehabilitation. The goal is to design a wearable device that uses sensors to detect hand movements and provide feedback to the user, allowing them to perform exercises to meet the specific needs of each individual stroke survivor.

II. Literature Review

The study by Shatil Imtiaz,Md Fardin Khan,Rahman Humyra,Md.Sajid Hossain,Md.Radsharar Kashem and Md. Humayun Kabir presents the development of a smart hand gloves for storke rehabilitation.The research highlights the growing need for affordable and accessible rehabilitation technologies for stroke survivors, particularly those experiencing motor impairments in their hands.

Existing studies emphasize the importance of sensor-based rehabilitation devices in assisting motor function recovery. The integration of flex sensors, force-sensitive resistors (FSRs), and accelerometers into wearable devices has shown promise in improving rehabilitation outcomes. Prior research has explored robotic exoskeletons, soft actuators, and virtual reality-based rehabilitation to enhance patient engagement and recovery. However, many existing technologies are either too expensive or complex for independent use.

This study aims to bridge that gap by introducing an ESP32-based smart glove that continuously monitors and records hand movements to assist both therapists and patients. Similar approaches in sensor-based rehabilitation, including EMG-controlled exoskeletons and force-feedback gloves, support the notion that real-time data collection can significantly improve rehabilitation programs.

The novelty of this research lies in integrating multiple sensor technologies into a cost-effective and user-friendly rehabilitation solution. The device enables continuous progress tracking, personalized rehabilitation exercises, and improved accessibility, contributing to the growing field of wearable healthcare technology.

The findings align with previous work on assistive rehabilitation devices and highlight the potential for home-based rehabilitation solutions. Future research could explore machine learning algorithms for adaptive therapy and wireless connectivity for remote monitoring, further enhancing the effectiveness of smart rehabilitation gloves.

III. Methodology

The development of the smart glove rehabilitation system follows a structured approach, integrating mirror therapy principles with sensor-based motion tracking and motor-assisted rehabilitation. The system consists of two gloves—one for the normal hand and the other for the paralyzed hand—that work in synchronization to facilitate movement recovery. The ESP32 microcontroller, along with flex sensors, force-sensitive resistors (FSRs), servo motors, and a motor driver (PCA9685), forms the core of the system.

1. System Design and Components

The rehabilitation system is designed with the following components:

1.1 Normal Hand Glove

Flex Sensor (Index Finger): Captures finger bending movements.

Force Sensitive Resistor (Palm): Detects grip pressure and hand activity.

ESP32: Processes sensor data and transmits it wirelessly to the paralyzed hand glove.

1.2 Paralyzed Hand Glove Servo Motors (MG996R):

Provide assisted movement for the affected fingers.

PCA9685 Motor Driver: Controls multiple servo motors.

ESP32: Receives sensor data from the normal hand glove and replicates the motion in the paralyzed hand.

2. Workflow and Implementation

The methodology is structured into two main stages:

2.1 Data Acquisition from the Normal Hand

1. The user wears the normal hand glove, which contains a flex sensor on the index finger and an FSR sensor on the palm.
2. When the index finger bends, the flex sensor records the bending angle.
3. When the user applies pressure on the palm**, the FSR sensor detects the force exerted.
4. The ESP32 microcontroller processes the data and wirelessly transmits it to the paralyzed hand glove.

2.2 Actuation of the Paralyzed Hand

1. The ESP32 on the paralyzed hand glove receives real-time sensor data from the normal hand glove.
2. Based on the received flex sensor data, the PCA9685 motor driver controls the servo motors attached to the paralyzed fingers.
3. The servo motors replicate the bending motion of the normal hand's index finger, assisting the corresponding finger in the paralyzed hand.
4. If grip pressure is detected in the normal hand's FSR sensor, an equivalent force is applied to the paralyzed hand by activating multiple servo motors in the glove.
5. The system continuously operates in real-time, reinforcing mirror therapy principles where the brain perceives movement in the affected hand by mimicking the healthy hand's actions.

3. Data Processing and Control Mechanism

-Calibration: The flex sensor is calibrated to map finger bending angles to servo motor positions.

Real-Time Synchronization: Wireless communication between the two ESP32 modules ensures minimal delay in movement replication.

Adaptive Assistance: The servo motor torque is adjusted based on the FSR sensor's pressure values, providing variable movement support.

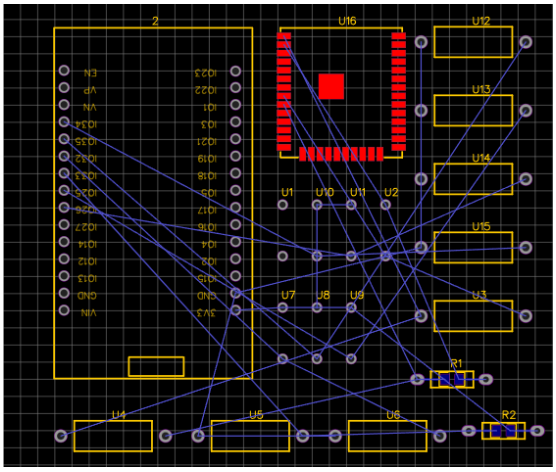


Fig. 1. PCB Layout

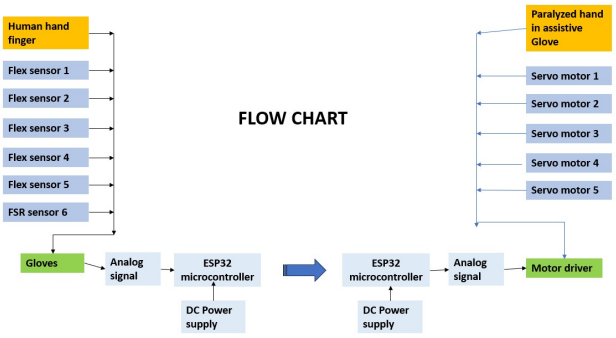


Fig. 2. Flow Chart

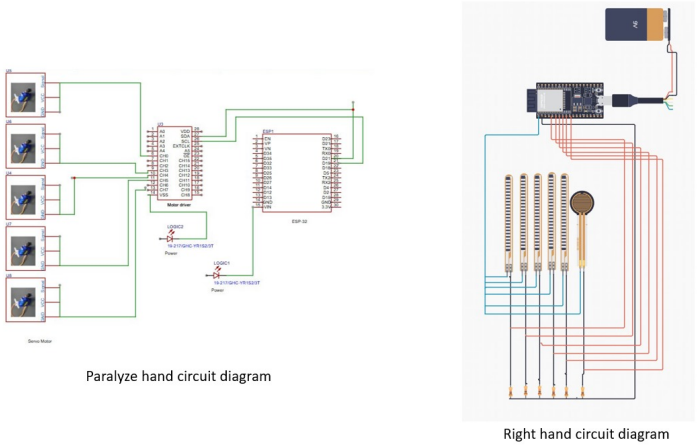


Fig. 3. Circuit Diagram

IV. Result

The developed smart glove rehabilitation system successfully integrates mirror therapy principles with sensor-based motion tracking and motor-assisted rehabilitation. The system

was tested to evaluate its performance in replicating normal hand movements in the paralyzed hand using real-time control. The results demonstrate the system's effectiveness in assisting stroke survivors with motor impairments.

1. Process Flow and Execution The experiment followed a structured process:

1.1 Motion Capture from the Normal Hand

The user wears the normal hand glove, which contains:
1.A flex sensor on the index finger to detect bending motion.
2.A force-sensitive resistor (FSR) on the palm to detect grip strength.
3.As the index finger bends, the flex sensor records the bending angle. When the user grips an object, the FSR sensor detects the applied force.
4.The ESP32 microcontroller processes this data and wirelessly transmits it to the paralyzed hand glove in real time.

1.2 Replication of Movement in the Paralyzed Hand

1.The ESP32 on the paralyzed hand glove receives the motion data from the normal hand.
2.The PCA9685 motor driver activates servo motors (MG996R) corresponding to each affected finger.
3.The servo motors rotate proportionally to the flex sensor values, replicating the bending movement of the normal hand's index finger.
4.When grip pressure is detected from the FSR sensor, the servo motors in the paralyzed hand close in response, simulating a natural grasp.
5.The process repeats continuously, creating a real-time mirror therapy effect.

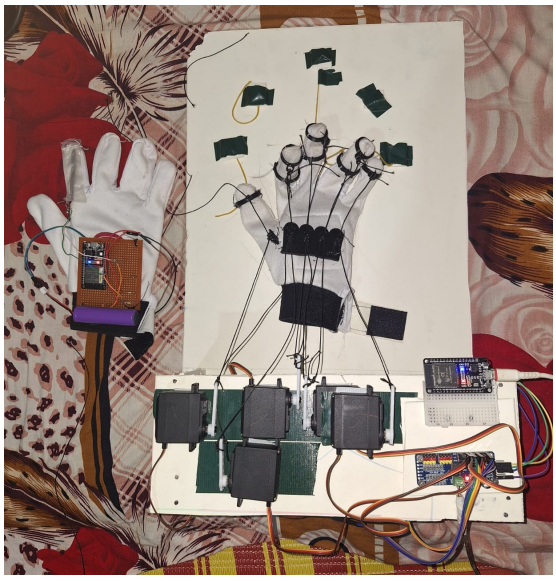


Fig. 4. Final Arrangment

V. Conclution

The development of the smart glove rehabilitation system successfully integrates mirror therapy principles with sensor-based motion tracking and motor-assisted movement to assist stroke survivors in hand rehabilitation. By using two gloves—one for the normal hand and one for the paralyzed hand—the system replicates natural hand motions in real time, allowing the affected hand to mimic the movements of the healthy hand. The combination of flex sensors, force-sensitive resistors (FSRs), servo motors, and an ESP32-based wireless system ensures smooth and synchronized operation.

The results demonstrated high accuracy in motion replication, with minimal latency in wireless communication. The servo motors effectively mimicked the bending motion of the normal hand's fingers, while the FSR sensor successfully triggered gripping actions in the paralyzed hand. Despite minor challenges such as power fluctuations, sensor interference, and servo jitter, these were mitigated through hardware optimization and software adjustments, leading to a functional and reliable rehabilitation device.

VI. References

[1] https://www.researchgate.net/publication/378782213 A Mirror-Hand Rehabilitation System Based on Virtual Reality Interaction

[2] https://ieeexplore.ieee.org/document/10441307

[3] https://pmc.ncbi.nlm.nih.gov/articles/PMC10295999/

[4] https://www.ijhsr.org/IJHSR Vol.14 Issue.5 May2024/IJHSR23.pdf

[5] https://dl.acm.org/doi/10.1145/2800835.2801648

[6] https://pmc.ncbi.nlm.nih.gov/articles/PMC6513639/